

Order-Dependent Triangular Limits and Double-Scaling Transitions for the Logarithmic Integral

Poisson Approximation, Borel Scaling, and a Shifted-Gaussian Phase

Independent research draft

Focused revision — August 30, 2026

This paper studies a triangular transformation whose outer series diverges for every fixed nonzero parameter has a convergent order-dependent specialization when the parameter is coupled to the truncation order by $a = N$. The limit is $\text{Li}(x) - \text{Li}(x/e)$ for $x > e$; an explicit global error estimate, the first two exact coefficients of its N^{-1} expansion, and a uniform remainder theorem at every fixed order are obtained. In the double-scaling regime $N(\log x - 1)^2 \rightarrow u^{-1}$, the normalized diagonal error has an exact Borel-model limit given by a positive-direction Borel sum and an equivalent Laplace-type integral; the same normalized model governs every fixed moving-window coupling $a = \lambda N$ at its corresponding threshold. When the window itself grows, so that $\lambda = \lambda_N \rightarrow 0$, a second theorem gives a δ -dependent shifted-Gaussian model and makes the $N \asymp (\log x)^2$ conditioning scale of the directly anchored member rigorous. Its large-scaling limit is finite, unlike the fixed-window Borel model, and its first finite-window correction is computed explicitly. An exact two-parameter identity and global squeeze show that $u\delta^2$ governs the small- δ crossover between the two scaling phases. Auxiliary enclosures, multiscale expansions, numerical diagnostics, and implementation formulas are collected in a separate supplementary note so that the article remains centered on this transition mechanism.

Abstract

Let $\text{Li}(x) = \int_2^x dt/\log t$ and $L = \log x$. We study the triangular finite expressions

$$T_N(x; a) = x \sum_{k=0}^N \sum_{m=0}^k \frac{(-1)^{k+m} k! (1-a)^{k-m}}{a^{k+1} (k-m)! L^{m+1}}.$$

For every fixed $a \neq 0$, the k th outer block is asymptotic to $x^a k!/(aL)^{k+1}$, so the associated outer series diverges. Nevertheless, coupling the parameter to the order gives

$$T_N(x; N) \longrightarrow \text{Li}(x) - \text{Li}(x/e) = x \sum_{m \geq 0} \frac{\gamma(m+1, 1)}{L^{m+1}}, \quad x > e.$$

The coefficients are both Riemann sums and scaled negative-binomial tails. These two interpretations yield an explicit global error bound and the sharper asymptotic formula

$$T_N(x; N) - (\text{Li}(x) - \text{Li}(x/e)) = \frac{C_1(x)}{N} + \frac{C_2(x)}{N^2} + O_x(N^{-3}),$$

where both C_1 and C_2 are given in closed form. At every fixed order J , a global likelihood-ratio estimate also gives a uniform remainder with the predicted near-threshold pole $(L-1)^{-2J-2}$. Thus the conclusion is an order-dependent limit, not the sum of a fixed-parameter series. In the near-threshold regime $L = 1 + \delta$ with $1/(N\delta^2) \rightarrow u \in (0, \infty)$, we further prove the Borel scaling law

$$\frac{e}{x} [T_N(x; N) - (\text{Li}(x) - \text{Li}(x/e))] \longrightarrow \int_0^\infty (e^{-u\rho^2/2} - 1) e^{-\rho} \frac{d\rho}{\rho},$$

and identify the limiting integral with the positive-direction Borel sum of the leading-pole error series. A Mellin representation also yields an exact convergent inverse-half-power expansion of this model for every positive scaling parameter. More generally, for every fixed $\lambda > 0$, the same Borel model governs the moving-window coupling $a = \lambda N$ when $\lambda^2 N (\log x - 1/\lambda)^2$ has a finite nonzero limit. For $\lambda_N \rightarrow 0$ with the absolute boundary gap $\delta = \log x - 1/\lambda_N$ fixed, the limiting shape changes to the shifted model

$$\Psi_\delta(u) = \int_0^\infty e^{-\rho} \frac{e^{-u(\rho-\delta)^2/2} - 1}{\rho - \delta} d\rho,$$

thereby proving the quadratic truncation scale for the member anchored at 2. It has an exact decomposition in terms of Ei , erfi , and two exponentially small boundary tails. For fixed $\delta > 0$ it tends to $e^{-\delta} \text{Ei}(\delta)$ as $u \rightarrow \infty$, showing that its recovery of the fixed-window Borel model as $\delta \downarrow 0$ is nonuniform. An exact crossover identity and a global two-sided squeeze show that $v = \delta\sqrt{u}$ governs the small- δ transition between the two boundary regimes; under the nearest-integer scaling its first correction is $K_\delta(u)/\mu + O(\mu^{-2})$ with an explicit positive integral K_δ . The construction is placed in the classical context of exponential-integral asymptotics, order-dependent mappings, Poisson approximation, Borel–Laplace summation, and Mellin inversion. The claims concern the precise limit, error, and transition theorems for the particular array studied here, rather than new general summability or special-function machinery.

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1 Introduction

The logarithmic integral is both elementary in definition and subtle in computation. Its standard large- x expansion,

$$\text{Li}(x) \sim \frac{x}{\log x} \sum_{k \geq 0} \frac{k!}{(\log x)^k}, \quad (1)$$

is asymptotic rather than convergent. This distinction matters whenever a finite transformation of the factorial coefficients appears numerically stable: good truncations do not, by themselves, define a convergent infinite series.

The question is whether a finite triangular transformation can converge when its fixed-parameter outer series does not. The answer is yes if the parameter changes with the truncation order. For the particular coupling $a = N$, the transformed coefficients become both positive Riemann sums and scaled negative-binomial tails converging to lower incomplete-gamma moments. These structures give a limit theorem and quantitative asymptotics. More importantly, the degeneration of those asymptotics near the convergence boundary produces a Borel scaling law, a shifted model when the window length grows, and an exact crossover between the two phases. This transition structure is the organizing subject of the article. Related but nonessential enclosure, multiscale, and numerical material is separated into the accompanying supplementary note.

1.1 Main contributions

The central results are as follows.

- Proposition 3.1 proves factorial divergence of the outer blocks for every fixed $a \neq 0$.
- Theorem 4.3 proves that $T_N(x; N)$ converges exactly for $x > e$, identifies its limit, and proves divergence to $+\infty$ for $1 < x \leq e$.
- Theorems 5.2 and 5.3 strengthen the coupled limit to a fully explicit $O_x(N^{-1})$ bound and a uniform bound with the correct threshold pole $(L - 1)^{-2}$.
- Theorem 5.6 determines the first two exact diagonal error coefficients and proves an $O_x(N^{-3})$ remainder after their subtraction.
- Theorem 5.13 proves a global remainder bound at every fixed order, with the correct near-threshold pole $(L - 1)^{-2J-2}$.
- Theorem 5.16 proves that, when $N(\log x - 1)^2 \rightarrow u^{-1}$, the normalized diagonal error converges to the Borel scaling model $\mathcal{B}\Phi(u)$, equivalently to the Laplace-type integral (41); Proposition 5.14 also gives its exact convergent expansion (42). Theorem 5.18 proves that this scaling model is invariant throughout the fixed moving-window family $a = \lambda N$, after the natural rescaling by λ . Theorem 5.19 treats the singular extension $\lambda_N \rightarrow 0$, identifies the resulting δ -dependent model, and proves the quadratic conditioning scale of the directly anchored family; Theorem 5.21, Proposition 5.22, Corollary 5.23, and Proposition 5.25 give its exact two-parameter crossover, special-function decomposition, distinct large- u limit, and first $1/\mu$ correction.

1.2 Normalization and domain

Throughout,

$$\text{Li}(x) := \int_2^x \frac{dt}{\log t}, \quad x > 1. \quad (2)$$

This is the offset logarithmic integral, not the polylogarithm. It is related to the principal-value convention by

$$\text{Li}(x) = \text{Ei}(\log x) - \text{Ei}(\log 2).$$

All difference identities are independent of the additive normalization.

2 Classical context and the boundary of the claim

The identities $\text{li}(x) = \text{Ei}(\log x)$, the power series for Ei , and its factorial asymptotic expansion are classical; convenient authoritative summaries are given in the NIST Digital Library of Mathematical Functions (DLMF) [1, 2, 3]. Incomplete gamma functions and their analytic conventions are likewise standard [4, 8]; general background on asymptotic expansions is given by Olver [9].

Convergent incomplete-gamma representations of exponential-integral differences also have direct classical precedents. Tricomi's multiplication theorem for the incomplete gamma function was revisited by Gautschi, Harris, and Temme, who derived convergent expansions connecting exponential integrals at different arguments and showed that an apparently new formula was a limiting case of Tricomi's 1950 identity [13, 14]. With their arguments specialized to $z = L$ and a multiplier $1 - \mu/L$, the same window difference $\text{Ei}(L) - \text{Ei}(L - \mu)$ is in view. Accordingly, no novelty is claimed here for the existence of an incomplete-gamma representation of that difference. The question addressed below is different: whether the prescribed finite triangular array $T_N(x; a)$, whose fixed-parameter outer series diverges, develops that window difference as an order-dependent limit, and with what sharp boundary and finite- N asymptotics. Euler transformations of divergent or slowly convergent expansions are also classical. The DLMF explicitly discusses Euler transformation in connection with the exponential-integral asymptotic series [7]. More generally, allowing a transformation parameter to depend on the truncation order belongs to the theory of order-dependent mappings developed by Seznec and Zinn-Justin [10] and reviewed by Zinn-Justin [11]. Consequently, neither a finite triangular rearrangement nor the general idea of order dependence is claimed here as a new method.

The analogy with order-dependent mapping (ODM) is structural rather than methodological. In classical ODM, an order-varying parameter is optimized within a resummation designed to recover the function represented asymptotically by a fixed divergent series. Here the prescribed diagonal $a = N$ is not selected by such an optimization, and its unit-window limit is the finite difference $\text{Li}(x) - \text{Li}(x/e)$ rather than $\text{Li}(x)$ itself; see Section 6 for the anchored moving-window member of the same family.

What is established in this paper is narrower and exactly testable: the specific double sum $T_N(x; a)$ has a divergent fixed- a outer series; the particular diagonal sequence $a = N$ tends to a lower incomplete-gamma expansion; and the convergence admits the explicit bound in Theorem 5.2. The multiscale construction is similarly presented as an exact reorganization of elementary integral identities. A claim of historical priority for the formulas would require a more extensive bibliographic search than is attempted here.

Several ingredients used below are classical. The identity

$$\sum_{j \geq 1} \frac{1}{j} \binom{2j}{j} w^j = -2 \log \frac{1 + \sqrt{1 - 4w}}{2}$$

is an elementary central-binomial generating-function identity; the Borel transform

$$\hat{\Phi}(t) = \log \frac{2}{1 + \sqrt{1 + 2t}}$$

is obtained from it by setting $w = -t/2$ and dividing by two. Thus no novelty is claimed for this closed form itself. What appears not to be identified in the sources reviewed is its occurrence as the Borel

transform governing the double-scaling diagonal error of the order-dependent triangular transformation for $\text{Li}(x)$.

This should also be separated from the classical erfc transition theory for incomplete gamma and beta functions. Temme's uniform expansions contain a complementary error function together with an asymptotic series and are uniformly valid in parameter domains [12]; modern transition-region expansions with explicitly computable polynomial coefficients were developed by Nemes and Olde Daalhuis [15]. For the generalized exponential integral, Olver's exponentially improved expansion and Dunster's error analysis likewise exhibit classical error-function transition structure [16, 17]. DLMF §8.18 records the corresponding incomplete-beta transition form [18]. These results are structurally adjacent, but they concern incomplete special functions in their own large-parameter transition regimes, whereas the present erfc /Borel model arises from the binomial-to-Poisson ratio inside the diagonal error.

The Poisson-approximation background is also classical. Uspensky-type expansions [19] and their reformulations via the Poisson convolution semigroup, including exact evaluations and remainder estimates in probability metrics, are treated by Deheuvels and Pfeifer [20] and generalize Shorgin's work [21]. Barbour obtained higher-order Poisson-limit expansions, including expansions for expectations of polynomially growing functions, while Roos developed sharp Poisson-binomial bounds using Charlier expansions [22, 23]. Analytic de-Poissonization supplies a separate, broad framework for transferring asymptotics from Poisson transforms to fixed-size models [24]. These results make the binomial-to-Poisson correction mechanism classical. They do not, however, directly state the factorially weighted triangular limit below: here the relevant object is a pointwise probability ratio inserted into a singular diagonal moment sum, rather than a direct application of an analytic de-Poissonization theorem.

This conservative positioning is mathematically useful. It separates three assertions that are often blurred in numerical experiments:

Object	Correct interpretation
Fixed- a outer sum	Divergent factorial series; finite truncations may still be useful.
Diagonal $a = N$ sequence	A distinct order-dependent sequence with a genuine limit for $x > e$.
Multiscale moment sum	A finite telescoping decomposition with convergent local series for $a > 1/2$.

3 The triangular family and fixed-parameter divergence

Let $x > 1$, write $L = \log x > 0$, and define, for $N \geq 0$ and $a \neq 0$,

$$T_N(x; a) := x \sum_{k=0}^N \sum_{m=0}^k \frac{(-1)^{k+m} k! (1-a)^{k-m}}{a^{k+1} (k-m)! L^{m+1}}. \quad (3)$$

The k th outer block is

$$B_k(x; a) := x \sum_{m=0}^k \frac{(-1)^{k+m} k! (1-a)^{k-m}}{a^{k+1} (k-m)! L^{m+1}}, \quad (4)$$

so that $T_N(x; a) = \sum_{k=0}^N B_k(x; a)$.

One way to view (3) is as a shifted endpoint expansion. Formally, after writing the upper-endpoint contribution to the exponential integral using $u = xe^{-t}$, so that $t = L - \log u$, one expands

$$\frac{1}{L-t} = \frac{1}{aL + \{(1-a)L - t\}} \sim \sum_{k \geq 0} \frac{(-1)^k ((1-a)L - t)^k}{(aL)^{k+1}}$$

and integrates the moments $\int_0^\infty t^m e^{-t} dt = m!$. Expanding the polynomial $((1-a)L-t)^k$ gives exactly the block $B_k(x; a)$. Thus the triangular family is not an unrelated double sum; it is the finite, shifted-anchor version of the usual endpoint asymptotic calculus for $\text{Ei}(L)$.

Introduce the truncated exponential polynomial

$$E_k(z) := \sum_{r=0}^k \frac{z^r}{r!}.$$

Proposition 3.1 (Exact block formula and fixed-parameter divergence). *For every fixed $x > 1$ and fixed real $a \neq 0$,*

$$B_k(x; a) = \frac{xk!}{(aL)^{k+1}} E_k(-(1-a)L), \quad (5)$$

and

$$B_k(x; a) \sim \frac{x^a k!}{(aL)^{k+1}} \quad \text{as } k \rightarrow \infty. \quad (6)$$

In particular $B_k(x; a)$ does not tend to zero, and $\sum_{k \geq 0} B_k(x; a)$ diverges.

Proof. Set $r = k - m$ in (4). Since $(-1)^{k+m} = (-1)^{2k-r} = (-1)^r$, one obtains

$$B_k(x; a) = \frac{xk!}{a^{k+1} L^{k+1}} \sum_{r=0}^k \frac{[-(1-a)L]^r}{r!},$$

which is (5). For fixed complex z , $E_k(z) \rightarrow e^z$. Here this limit is $e^{-(1-a)L} = x^{a-1}$, which is nonzero. This proves (6). Moreover, the magnitude ratio of the asymptotic comparison terms is $(k+1)/|aL|$, which tends to infinity. Thus the terms cannot tend to zero and the outer series diverges. $\square$

Remark 3.2. Proposition 3.1 rules out the most tempting but incorrect proof strategy: one cannot first sum the fixed- a outer series and then substitute $a = N$. The symbol N would then simultaneously mean a completed infinite sum and a changing parameter. The diagonal sequence must instead be analyzed directly as a sequence of finite sums. The fixed- a divergence is a large- k phenomenon, becoming visible once k is much larger than the natural scale $|aL|$. The diagonal $a = N$ keeps the summation range $k \leq N = a$ on this moving pre-asymptotic scale, which is the elementary mechanism behind the otherwise surprising stabilization.

Figure 1 separates finite-order usefulness from convergence. Before the factorial regime dominates, a fixed- a truncation can be numerically accurate; adding further terms eventually destroys that accuracy. This is the characteristic behavior of an asymptotic expansion and is why a plot of only the first few orders cannot establish convergence.

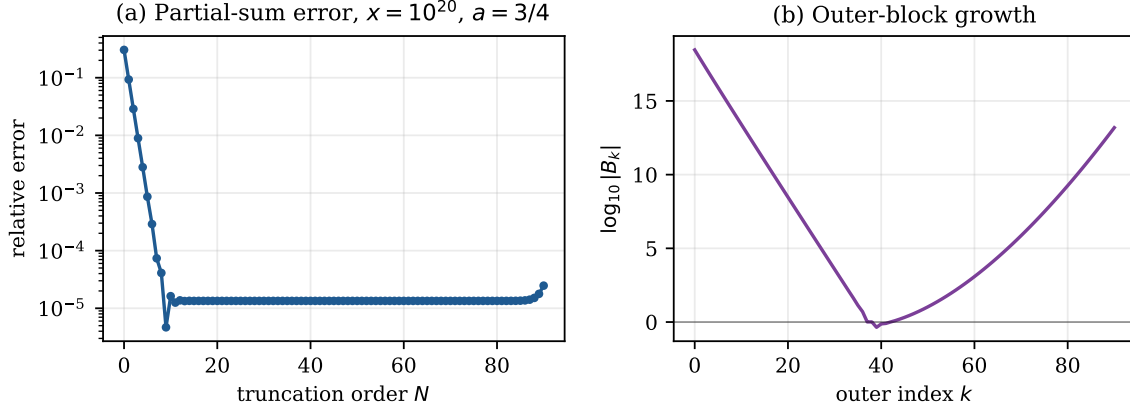


Figure 1: Fixed-parameter truncations at $x = 10^{20}$ and $a = 3/4$. Left: the relative error of $T_N(x; a)$ against the associated window integral initially decreases, reaches an optimal finite truncation, and then grows without bound. Right: the same transition is visible directly in the outer blocks, whose eventual factorial growth is proved in Proposition 3.1.

4 The coupled choice $a = N$

The cancellation that occurs after setting $a = N$ is most transparent after reversing the order of summation. For $0 \leq m \leq N$, define

$$c_{N,m} := \frac{1}{N} \sum_{j=0}^{N-m} \left(\prod_{r=1}^m \frac{j+r}{N} \right) \left(1 - \frac{1}{N} \right)^j, \quad (7)$$

where an empty product is one. Set $c_{N,m} = 0$ for $m > N$.

Lemma 4.1 (Positive coefficient representation). *For every $N \geq 1$,*

$$\frac{T_N(x; N)}{x} = \sum_{m=0}^N \frac{c_{N,m}}{L^{m+1}}, \quad (8)$$

and $0 \leq c_{N,m} \leq 1$.

Proof. In (3) set $k = m + j$. The two signs from $(-1)^j$ and $(1 - N)^j = (-1)^j(N - 1)^j$ cancel. Also

$$\frac{(m+j)!}{j!} \frac{(N-1)^j}{N^{m+j+1}} = \frac{1}{N} \left(\prod_{r=1}^m \frac{j+r}{N} \right) \left(1 - \frac{1}{N} \right)^j.$$

Summing first in j gives (8). For $0 \leq j \leq N - m$ and $1 \leq r \leq m$, one has $j + r \leq N$, so every product factor lies in $[0, 1]$. Consequently

$$0 \leq c_{N,m} \leq \frac{1}{N} \sum_{j=0}^{N-m} \left(1 - \frac{1}{N} \right)^j \leq \frac{1}{N} \sum_{j=0}^{\infty} \left(1 - \frac{1}{N} \right)^j = 1.$$

□

Lemma 4.2 (Coefficient limit). *For every fixed integer $m \geq 0$,*

$$c_{N,m} \longrightarrow \gamma(m+1, 1) := \int_0^1 t^m e^{-t} dt. \quad (9)$$

Proof. For fixed m , write $t_j = j/N$. Uniformly for $0 \leq j \leq N - m$,

$$\prod_{r=1}^m \frac{j+r}{N} = \prod_{r=1}^m (t_j + r/N) \longrightarrow t_j^m$$

in the Riemann-sum sense, while $(1 - 1/N)^j \rightarrow e^{-t_j}$ uniformly on $0 \leq j \leq N$. The missing terminal interval has length $m/N \rightarrow 0$. Thus (7) is a Riemann sum for the integral in (9). $\square$

Theorem 4.3 (Coupled triangular limit). *Let $x > 1$ and $L = \log x$. If $x > e$, then*

$$T_N(x; N) \longrightarrow D(x) := \text{Li}(x) - \text{Li}(x/e) = x \sum_{m=0}^{\infty} \frac{\gamma(m+1, 1)}{L^{m+1}} \quad (10)$$

and the series on the right is absolutely convergent. If $1 < x \leq e$, then $T_N(x; N) \rightarrow +\infty$. Thus the diagonal sequence has a finite limit exactly for $x > e$.

Proof. Assume first that $L > 1$. Lemma 4.1 gives

$$0 \leq \frac{c_{N,m}}{L^{m+1}} \leq \frac{1}{L^{m+1}},$$

and the majorant is summable. Extend $c_{N,m}$ by zero for $m > N$. Dominated convergence for series, together with Lemma 4.2, yields

$$\frac{T_N(x; N)}{x} \longrightarrow \sum_{m=0}^{\infty} \frac{\gamma(m+1, 1)}{L^{m+1}}.$$

All terms are nonnegative, so the limiting series is absolutely convergent.

It remains to identify the sum. A change of variables $u = xe^{-t}$ gives

$$\text{Li}(x) - \text{Li}(x/e) = x \int_0^1 \frac{e^{-t}}{L - t} dt. \quad (11)$$

Because $0 \leq t/L \leq 1/L < 1$, the uniformly convergent geometric expansion

$$\frac{1}{L - t} = \sum_{m=0}^{\infty} \frac{t^m}{L^{m+1}}$$

may be integrated termwise in (11). This proves the identity in (10).

Now let $0 < L \leq 1$. For every fixed M , positivity and the coefficient limit imply

$$\liminf_{N \rightarrow \infty} \frac{T_N(x; N)}{x} \geq \sum_{m=0}^M \frac{\gamma(m+1, 1)}{L^{m+1}}.$$

The endpoint-moment estimate

$$\gamma(m+1, 1) = \int_0^1 t^m e^{-t} dt \sim \frac{e^{-1}}{m+1} \quad (12)$$

shows that these partial sums diverge as $M \rightarrow \infty$: harmonically when $L = 1$ and faster when $L < 1$. Therefore $T_N(x; N) \rightarrow +\infty$. $\square$

Remark 4.4 (Meaning of “absolutely”). For each N , $T_N(x; N)$ is finite. Absolute convergence in Theorem 4.3 refers to the limiting incomplete-gamma series, while convergence of $T_N(x; N)$ refers to a sequence of different finite triangular sums. It does not repair the fixed- a series of Proposition 3.1.

Proposition 4.5 (A moving-window family). *Let $\lambda > 0$ be fixed and put $a = \lambda N$. If $L > 1/\lambda$, then*

$$T_N(x; \lambda N) \longrightarrow x \sum_{m=0}^{\infty} \frac{\gamma(m+1, 1/\lambda)}{L^{m+1}} = \text{Li}(x) - \text{Li}(xe^{-1/\lambda}). \quad (13)$$

If $0 < L \leq 1/\lambda$, then $T_N(x; \lambda N) \rightarrow +\infty$.

Proof. Repeating the reindexing in Lemma 4.1 gives

$$\frac{T_N(x; \lambda N)}{x} = \sum_{m=0}^N \frac{c_{N,m}^{(\lambda)}}{L^{m+1}},$$

where

$$c_{N,m}^{(\lambda)} := \frac{1}{\lambda N} \sum_{j=0}^{N-m} \left(\prod_{r=1}^m \frac{j+r}{\lambda N} \right) \left(1 - \frac{1}{\lambda N} \right)^j.$$

For fixed m , with $t_j = j/(\lambda N)$, this is a Riemann sum on $0 \leq t \leq 1/\lambda$, hence

$$c_{N,m}^{(\lambda)} \longrightarrow \int_0^{1/\lambda} t^m e^{-t} dt = \gamma(m+1, 1/\lambda).$$

Also $j+r \leq N$ on the summation range, so each product factor is at most $1/\lambda$, and

$$0 \leq c_{N,m}^{(\lambda)} \leq \lambda^{-m} \frac{1}{\lambda N} \sum_{j \geq 0} \left(1 - \frac{1}{\lambda N} \right)^j = \lambda^{-m}$$

for all sufficiently large N . Therefore $c_{N,m}^{(\lambda)}/L^{m+1}$ is dominated by λ^{-m}/L^{m+1} , which is summable exactly when $\lambda L > 1$. Dominated convergence proves the series limit. Finally,

$$\text{Li}(x) - \text{Li}(xe^{-1/\lambda}) = x \int_0^{1/\lambda} \frac{e^{-t}}{L-t} dt,$$

and termwise integration of $(L-t)^{-1}$ identifies the sum when $L > 1/\lambda$.

If $0 < L \leq 1/\lambda$, positivity gives the same lower-bound argument as in Theorem 4.3. Since

$$\gamma(m+1, 1/\lambda) = \int_0^{1/\lambda} t^m e^{-t} dt \sim \frac{e^{-1/\lambda} (1/\lambda)^{m+1}}{m+1},$$

the limiting partial sums diverge harmonically at $L = 1/\lambda$ and faster below it. $\square$

Remark 4.6. Proposition 4.5 shows that the diagonal $a = N$ is the unit-window member of a larger moving-window family. The convergence boundary is not accidental: it is exactly the condition that the logarithmic window length $1/\lambda$ stay below $L = \log x$. The double-scaling theorem below is first proved for the unit case $\lambda = 1$ and then extended to the whole λ -family in Theorem 5.18.

The three panels in Figure 2 show what the limiting statements mean as functions of x , not only at isolated test points. The incomplete-gamma truncations improve geometrically, whereas the triangular diagonal approaches the same unit window only at an algebraic rate. Allowing the window length to depend on x supplies the normalization $\text{Li}(2) = 0$ and makes the right-hand panel a direct approximation to $\text{Li}(x)$.

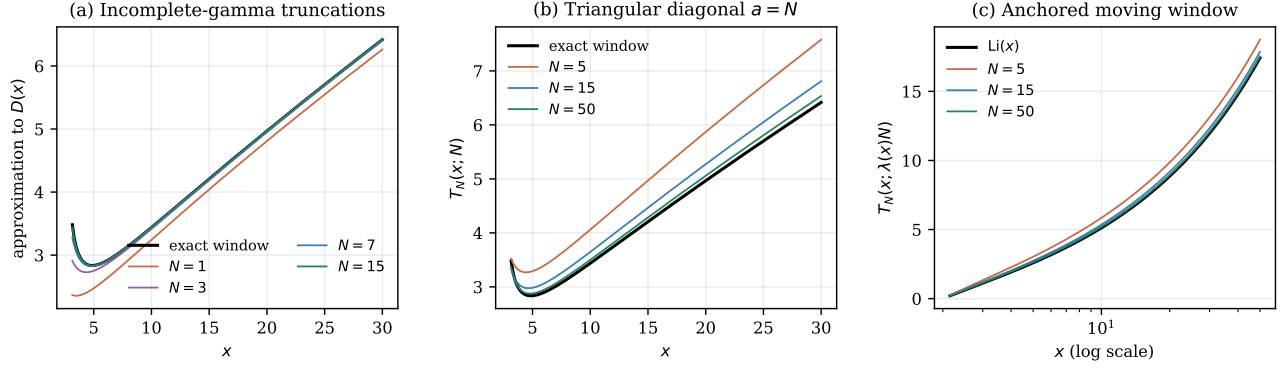


Figure 2: Representative truncation curves for the convergent constructions. Left: the positive lower incomplete-gamma series for the unit window $D(x)$. Center: the order-dependent diagonal $T_N(x; N)$ approaching that window. Right: the anchored choice $\lambda(x) = 1/(\log x - \log 2)$, for which $T_N(x; \lambda(x)N)$ approaches $\text{Li}(x)$ directly. Black curves are reference integrals and colored curves are the displayed truncation orders.

Proposition 4.7 (Distributional identity). *Let S_{m+1} be the number of Bernoulli($1/N$) trials required to obtain the $(m+1)$ st success, and let $X_N \sim \text{Bin}(N+1, 1/N)$. For $0 \leq m \leq N$,*

$$c_{N,m} = m! \Pr\{S_{m+1} \leq N+1\} = m! \Pr\{X_N \geq m+1\}. \quad (14)$$

Moreover, if $Y \sim \text{Poisson}(1)$, then

$$\gamma(m+1, 1) = m! \Pr\{Y \geq m+1\}. \quad (15)$$

Equivalently, the binomial-tail identity has the standard incomplete-beta form

$$\Pr\{X_N \geq m+1\} = I_{1/N}(m+1, N-m+1), \quad (16)$$

where $I_p(a, b)$ is the regularized incomplete beta function [5].

Proof. Writing $p = 1/N$ and $q = 1 - p$, equation (7) becomes

$$c_{N,m} = m! \sum_{j=0}^{N-m} \binom{m+j}{j} p^{m+1} q^j.$$

The summand is the probability that the $(m+1)$ st success occurs on trial $m+1+j$; summing through $j = N-m$ gives the first probability in (14). That waiting-time event is equivalent to obtaining at least $m+1$ successes among the first $N+1$ trials, proving the second identity. Finally, the integer-order incomplete-gamma identity

$$\frac{\gamma(m+1, 1)}{m!} = 1 - e^{-1} \sum_{r=0}^m \frac{1}{r!} = \Pr\{Y \geq m+1\}$$

proves (15). $\square$

Remark 4.8. The coupling $a = N$ therefore has a distributional interpretation: a discrete waiting-time law with success probability $1/N$ is compared on the time scale N with its continuous gamma limit. This explains why the signed double sum becomes positive after the diagonal substitution. The elementary bound $c_{N,m} \leq 1$ used in Theorem 4.3 remains stronger than the immediate probability bound $c_{N,m} \leq m!$.

Lemma 4.9 (Global likelihood-ratio estimate). *Let $p_N(r) = \Pr\{X_N = r\}$, $p(r) = \Pr\{Y = r\} = e^{-1}/r!$, and put $\rho_N(r) = p_N(r)/p(r)$ for $0 \leq r \leq N+1$ and $\rho_N(r) = 0$ otherwise. There is an absolute constant $B > 0$ such that, for every $N \geq 2$ and $r \geq 1$,*

$$|\rho_N(r) - 1| \leq B \min\left(1, \frac{r^2}{N}\right). \quad (17)$$

Proof. For $0 \leq r \leq N+1$ the exact ratio is

$$\rho_N(r) = e^{\frac{(N+1)^r}{Nr}} \left(1 - \frac{1}{N}\right)^{N+1-r}.$$

The falling product and the last binomial factor give $p_N(r) \leq (1 + 1/N)/r!$, hence $\rho_N(r) \leq e(1 + 1/N)$ and $|\rho_N(r) - 1| \leq 4$. When $1 \leq r \leq N/2$ and $r^2 \leq N$, expand the logarithm of the displayed ratio. The elementary estimate $\log(1 + y) = y - y^2/2 + O(|y|^3)$, uniformly for $|y| \leq 1/2$, gives

$$|\log \rho_N(r)| \leq B_0 \frac{r^2}{N}$$

with an absolute B_0 . Since $r^2/N \leq 1$, the inequality $|e^v - 1| \leq e^{|v|}|v|$ gives the second branch of (17). All remaining indices are covered by the global bound after increasing B ; the finitely many small values of N are absorbed in the same constant. $\square$

5 A quantitative $O_x(N^{-1})$ theorem

The Riemann-sum proof above can be made uniform in the coefficient index. This gives a practical error estimate and shows that the observed stabilization is genuinely first order in N .

Lemma 5.1 (Uniform coefficient estimate). *For $N \geq 2$ and every $m \geq 0$,*

$$|c_{N,m} - \gamma(m+1, 1)| \leq \frac{m^2 + 2m + 2}{N} + \frac{1}{N-1}. \quad (18)$$

Proof. It is enough first to consider $0 \leq m \leq N$. Put $t_j = j/N$ and

$$P_{N,m}(j) := \prod_{r=1}^m (t_j + r/N).$$

For $0 \leq j \leq N-m$, all factors lie in $[0, 1]$. A telescoping product estimate therefore gives

$$|P_{N,m}(j) - t_j^m| \leq \sum_{r=1}^m \frac{r}{N} = \frac{m(m+1)}{2N}. \quad (19)$$

For $m \geq 1$, the inequality $m+1 \leq 2m$ gives the simpler upper bound m^2/N . For $m = 0$, both products are empty and the difference is exactly zero.

Next, for $N \geq 2$,

$$-\frac{1}{N-1} \leq \log(1 - 1/N) \leq -\frac{1}{N}.$$

For $0 \leq j \leq N$ this implies

$$0 \leq e^{-j/N} - (1 - 1/N)^j \leq e^{-j/N} \frac{j}{N(N-1)} \leq \frac{1}{N-1}. \quad (20)$$

The first weight estimate follows from $1 - e^{-u} \leq u$; the last follows from $j \leq N$. If $m \geq 1$, the sum contains at most N terms. Since $t_j^m \leq 1$, averaging (19) and (20) then changes $c_{N,m}$ by at most $m^2/N + 1/(N-1)$ from the truncated Riemann sum

$$\frac{1}{N} \sum_{j=0}^{N-m} t_j^m e^{-t_j}.$$

If $m = 0$, the product error vanishes. Moreover $j e^{-j/N} \leq N e^{-1}$ on $0 \leq j \leq N$, so the sharper middle bound in (20) gives a total weight error at most

$$\frac{N+1}{N} \frac{e^{-1}}{N-1} < \frac{1}{N-1}.$$

Thus the same comparison bound holds in the exceptional $m = 0$ case.

For $f_m(t) = t^m e^{-t}$ on $[0, 1]$, one has $0 \leq f_m \leq 1$ and $|f'_m| \leq m + 1$. Comparing the left Riemann sum on the full grid with the integral, and then accounting for the at most $m + 1$ omitted or endpoint cells, gives the deliberately non-optimal but uniform estimate

$$\left| \frac{1}{N} \sum_{j=0}^{N-m} f_m(t_j) - \int_0^1 f_m(t) dt \right| \leq \frac{2m+2}{N}. \quad (21)$$

Combining the three bounds proves (18) for $m \leq N$. If $m > N$, then $c_{N,m} = 0$ and $0 \leq \gamma(m+1, 1) \leq 1$, whereas the right side of (18) exceeds one. Hence the estimate holds for all m . $\square$

Theorem 5.2 (Explicit convergence rate). *Let $x > e$, $L = \log x$, and $N \geq 2$. Then*

$$|T_N(x; N) - D(x)| \leq x \left[\frac{L+1}{N(L-1)^3} + \frac{2}{N(L-1)^2} + \frac{2}{N(L-1)} + \frac{1}{(N-1)(L-1)} \right]. \quad (22)$$

In particular,

$$T_N(x; N) = \text{Li}(x) - \text{Li}(x/e) + O_x(N^{-1}).$$

Proof. By Theorem 4.3 and the zero extension of $c_{N,m}$,

$$|T_N(x; N) - D(x)| \leq x \sum_{m=0}^{\infty} \frac{|c_{N,m} - \gamma(m+1, 1)|}{L^{m+1}}.$$

Apply Lemma 5.1. The required sums are

$$\begin{aligned} \sum_{m \geq 0} \frac{1}{L^{m+1}} &= \frac{1}{L-1}, \\ \sum_{m \geq 0} \frac{m}{L^{m+1}} &= \frac{1}{(L-1)^2}, \\ \sum_{m \geq 0} \frac{m^2}{L^{m+1}} &= \frac{L+1}{(L-1)^3}. \end{aligned}$$

Substitution gives (22). $\square$

Theorem 5.3 (Uniform rate with the correct threshold pole). *There is an absolute constant $A > 0$ such that, for all $N \geq 2$ and $m \geq 0$,*

$$|c_{N,m} - \gamma(m+1, 1)| \leq \frac{A(1+m)}{N}. \quad (23)$$

Consequently, for every $x > e$, with $L = \log x$,

$$\boxed{|T_N(x; N) - D(x)| \leq \frac{AxL}{N(L-1)^2}}. \quad (24)$$

Thus a global coefficient estimate already has the same pole order as the exact first diagonal coefficient.

Proof. The probability-tail identities in Proposition 4.7 give

$$c_{N,m} - \gamma(m+1, 1) = m! \sum_{r>m} p(r)(\rho_N(r) - 1).$$

Lemma 4.9 and $\min(1, r^2/N) \leq r^2/N$ therefore imply

$$|c_{N,m} - \gamma(m+1, 1)| \leq \frac{B}{N} m! \sum_{r>m} p(r)r^2.$$

For $m \geq 1$, using $r^2 = r(r-1) + r$ gives

$$m! \sum_{r>m} p(r)r^2 = \frac{m!}{e} \left(\sum_{k=m-1}^{\infty} \frac{1}{k!} + \sum_{k=m}^{\infty} \frac{1}{k!} \right) = O(m+1)$$

with an absolute implied constant; the case $m = 0$ is immediate from $\mathbb{E}Y^2 = 2$. This proves (23). Finally,

$$\sum_{m \geq 0} \frac{m+1}{L^{m+1}} = \frac{L}{(L-1)^2},$$

which gives (24). □

Remark 5.4 (The candidate sharp constant). At $m = 0$ one has

$$c_{N,0} = 1 - \left(1 - \frac{1}{N}\right)^{N+1}, \quad N|c_{N,0} - \gamma(1, 1)| \rightarrow \frac{3}{2e}.$$

Thus no constant smaller than $3/(2e)$ can hold in (23). High-precision computations through $N = 400$ indicate that the maximum over m occurs at $m = 0$ and increases to $3/(2e)$, suggesting that this lower bound is the sharp constant. The proof above deliberately takes absolute values before summing and does not retain the cancellations needed to establish that sharper assertion.

Remark 5.5 (Sharpness and conditioning). The constant in (22) is chosen for transparency, not sharpness. It becomes large as $L \downarrow 1$, reflecting the genuine loss of summability at $x = e$. For any compact set $x \in [e^{1+\delta}, X]$, $\delta > 0$, the estimate is uniform after taking the maximum of its explicit prefactor.

5.1 The first two exact diagonal error coefficients

The probability representation in Proposition 4.7 permits a sharper result without requiring a coefficient expansion uniform in m .

Theorem 5.6 (First two diagonal error coefficients). *For every fixed $x > e$, with $L = \log x$,*

$$\boxed{T_N(x; N) - D(x) = \frac{C_1(x)}{N} + \frac{C_2(x)}{N^2} + O_x(N^{-3})}, \quad (25)$$

where

$$\boxed{C_1(x) = \frac{x(3L-4)}{2e(L-1)^2}, \quad C_2(x) = -\frac{x(7L^3 - 38L^2 + 81L - 68)}{24e(L-1)^4}}. \quad (26)$$

Proof. Put $z = 1/L \in (0, 1)$, let $X_N \sim \text{Bin}(N + 1, 1/N)$, and let $Y \sim \text{Poisson}(1)$. Define

$$h_z(0) = 0, \quad h_z(r) = \sum_{m=0}^{r-1} m! z^{m+1} \quad (r \geq 1).$$

Proposition 4.7, followed by an interchange of nonnegative sums, gives the exact expectation identities

$$\frac{T_N(x; N)}{x} = \mathbb{E}h_z(X_N), \quad \frac{D(x)}{x} = \mathbb{E}h_z(Y). \quad (27)$$

Write $p_N(r) = \Pr\{X_N = r\}$ and $p(r) = e^{-1}/r!$. For $0 \leq r \leq N + 1$, their ratio is

$$\frac{p_N(r)}{p(r)} = e^{\frac{(N+1)^r}{N^r}} \left(1 - \frac{1}{N}\right)^{N+1-r}, \quad (28)$$

where $u^k := u(u-1) \cdots (u-k+1)$ denotes a falling factorial. Taylor expansion of the logarithm in (28) gives, uniformly for $r \leq K \log N$ with fixed K ,

$$\log \frac{p_N(r)}{p(r)} = \frac{\alpha_r}{N} + \frac{\lambda_r}{N^2} + O_K \left(\frac{(1+r)^5}{N^3} \right), \quad (29)$$

$$\lambda_r = -\frac{r^3}{6} + \frac{3r^2}{4} - \frac{7r}{12} - \frac{5}{6}. \quad (30)$$

Exponentiating yields

$$\frac{p_N(r)}{p(r)} = 1 + \frac{\alpha_r}{N} + \frac{\beta_r}{N^2} + O_K \left(\frac{(1+r)^6}{N^3} \right), \quad (31)$$

where

$$\alpha_r = \frac{-r^2 + 5r - 3}{2}, \quad (32)$$

$$\beta_r = \lambda_r + \frac{1}{2}\alpha_r^2 = \frac{r^4}{8} - \frac{17r^3}{12} + \frac{37r^2}{8} - \frac{13r}{3} + \frac{7}{24}. \quad (33)$$

Indeed, the coefficient of N^{-1} in the logarithm of the ratio is

$$\sum_{s=0}^{r-1} (1-s) + (r-1) - \frac{1}{2} = \frac{-r^2 + 5r - 3}{2},$$

while the coefficient of N^{-2} is

$$-\frac{1}{2} \sum_{s=0}^{r-1} (1-s)^2 + \frac{r-1}{2} - \frac{1}{3} = \lambda_r.$$

The remaining logarithmic terms are $O_K((1+r)^5/N^3)$, and exponentiation gives (31).

The normalization of the probability masses supplies a useful consistency check. In falling factorials,

$$\alpha_r = -\frac{1}{2}r^2 + 2r^1 - \frac{3}{2},$$

$$\beta_r = \frac{1}{8}r^4 - \frac{2}{3}r^3 + \frac{5}{4}r^2 - r^1 + \frac{7}{24}.$$

For $Y \sim \text{Poisson}(1)$, one has $\mathbb{E}[Y^k] = 1$. Thus the coefficients in each line sum to zero and hence $\mathbb{E}\alpha_Y = \mathbb{E}\beta_Y = 0$. This is exactly what $\sum_r p_N(r) = \sum_r p(r) = 1$ forces at the first two orders.

For completeness, the expansion may be summed against h_z uniformly. There is a constant K_z such that

$$h_z(r) \leq K_z(r-1)!z^r \quad (r \geq 1),$$

because the terms $m!z^{m+1}$ are eventually geometrically increasing in m , and the finitely many preceding cases may be absorbed into K_z . The constant necessarily deteriorates as $z \uparrow 1$, which is one source of the x -dependence in the final remainder. Directly from the binomial mass function,

$$p_N(r) \leq \frac{(N+1)^r}{N^r r!} \leq \frac{1+1/N}{r!},$$

while $p(r) = e^{-1}/r!$. Hence both $p_N(r)h_z(r)$ and $p(r)h_z(r)$ are bounded by a constant times z^r/r , uniformly in N . Split the expectation at $R_N = \lceil K \log N \rceil$, choosing K so large that $z^{R_N} = O(N^{-4})$. On $r \leq R_N$, (31) has a summable error $O_z(N^{-3})$, because $\sum_{r \geq 1} (1+r)^6 z^r/r < \infty$ for $z < 1$. Beyond R_N , the preceding geometric majorant makes the full tail $O_z(N^{-4})$, even after the polynomial factors in α_r and β_r are included. Therefore

$$\mathbb{E}h_z(X_N) - \mathbb{E}h_z(Y) = \frac{1}{N} \mathbb{E}[\alpha_Y h_z(Y)] + \frac{1}{N^2} \mathbb{E}[\beta_Y h_z(Y)] + O_z(N^{-3}). \quad (34)$$

It remains to evaluate the expectation. The Poisson identities $rp(r) = p(r-1)$ and $r(r-1)p(r) = p(r-2)$ imply, for every $m \geq 0$,

$$\sum_{r=m+1}^{\infty} p(r)\alpha_r = \frac{3-m}{2e m!}, \quad (35)$$

$$\sum_{r=m+1}^{\infty} p(r)\beta_r = \frac{3m^3 - 22m^2 + 36m - 7}{24e m!}. \quad (36)$$

These follow immediately from the falling-factorial decompositions above, with absent negative-index Poisson masses interpreted as zero. Absolute convergence permits another interchange, giving

$$\begin{aligned} \mathbb{E}[\alpha_Y h_z(Y)] &= \sum_{m=0}^{\infty} m!z^{m+1} \sum_{r=m+1}^{\infty} p(r)\alpha_r \\ &= \frac{1}{2e} \sum_{m=0}^{\infty} (3-m)z^{m+1} = \frac{z(3-4z)}{2e(1-z)^2}. \end{aligned}$$

Similarly,

$$\begin{aligned} \mathbb{E}[\beta_Y h_z(Y)] &= \frac{1}{24e} \sum_{m=0}^{\infty} (3m^3 - 22m^2 + 36m - 7)z^{m+1} \\ &= \frac{z(-7 + 38z - 81z^2 + 68z^3)}{24e(1-z)^4}. \end{aligned}$$

Combining these evaluations with (27) and (34), then substituting $z = 1/L$, proves (25)–(26). $\square$

Corollary 5.7 (Sign, conditioning, and the large- x limit). *For all sufficiently large N , the diagonal approximation underestimates $D(x)$ when $e < x < e^{4/3}$ and overestimates it when $x > e^{4/3}$. At the transition point,*

$$C_1(e^{4/3}) = 0, \quad C_2(e^{4/3}) = 37e^{1/3} > 0,$$

so

$$T_N(e^{4/3}; N) - D(e^{4/3}) \sim \frac{37e^{1/3}}{N^2}.$$

Thus the approximation is an overestimate there as well, with error exactly of order N^{-2} . This is an asymptotic statement rather than a small- N estimate: because $L - 1 = 1/3$, the higher coefficients are strongly amplified, and the leading-pole estimate makes the relative third-order correction of order $30/N$.

Thus moderate values of N can remain visibly below the limiting scaled constant $37e^{1/3}$; see Remark 5.15. Moreover,

$$C_1(x) \sim -\frac{x}{2e(L-1)^2} \quad (L \downarrow 1), \quad \frac{C_1(x)}{D(x)} \longrightarrow \frac{3}{2(e-1)} \quad (x \rightarrow \infty).$$

Thus the leading relative error at large x is approximately $3/[2(e-1)N] \approx 0.873/N$.

Remark 5.8. The explicit bound in Theorem 5.2 grows like $(L-1)^{-3}$ near the threshold, whereas the true leading constant grows like $(L-1)^{-2}$. The quadratic m^2 term in Lemma 5.1 is therefore genuinely lossy; the exact first correction has only linear coefficient growth, $(3-m)/(2e)$. The second coefficient satisfies

$$C_2(x) \sim \frac{3x}{4e(L-1)^4} \quad (L \downarrow 1).$$

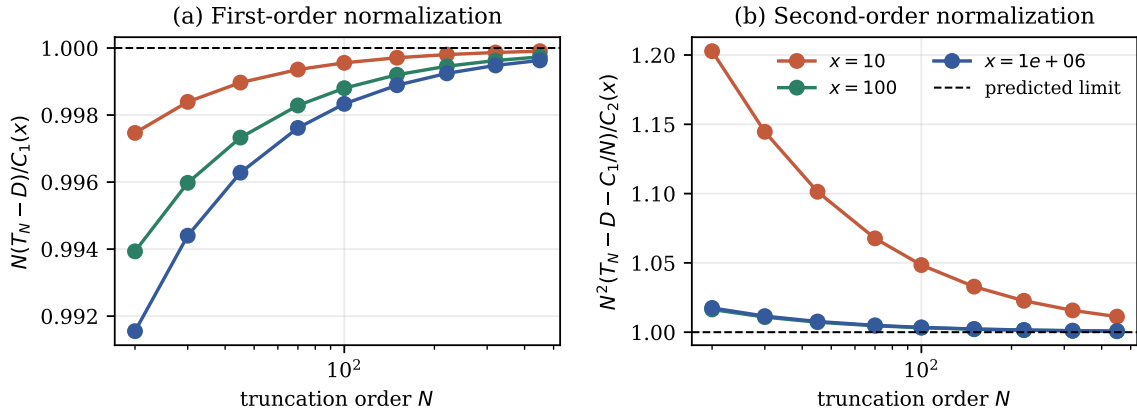


Figure 3: Normalized diagonal residuals for three values of x . The left panel divides the error by its first-order prediction $C_1(x)/N$; the right panel subtracts that contribution and divides by $C_2(x)/N^2$. The dashed line is the limiting value one in Theorem 5.6.

Figure 3 makes both orders of Theorem 5.6 visible. The first normalization already clusters close to one, while the second resolves the smaller correction left after subtracting C_1/N . The slower approach in the more strongly conditioned cases is consistent with the poles of C_1 and C_2 as $L \downarrow 1$.

Lemma 5.9 (Normalization of the ratio coefficients). *Let*

$$\rho_N(r) = \frac{p_N(r)}{p(r)}$$

be the probability ratio in (28), and suppose that, on $r \leq R_N = \lceil K \log N \rceil$,

$$\rho_N(r) = 1 + \sum_{j=1}^J \frac{A_j(r)}{N^j} + O_{J,K} \left(\frac{(1+r)^{2J+2}}{N^{J+1}} \right),$$

where each A_j is the polynomial coefficient obtained from the expansion of (28). Then

$$\mathbb{E}A_j(Y) = 0, \quad j = 1, \dots, J,$$

for $Y \sim \text{Poisson}(1)$.

Proof. For every N ,

$$\sum_{r \geq 0} p(r)(\rho_N(r) - 1) = \sum_{r=0}^{N+1} p_N(r) - 1 = 0.$$

Split the sum at $R_N = \lceil K \log N \rceil$. On $r \leq R_N$, the displayed expansion may be summed term by term because the Poisson distribution has finite moments of every order. The summed remainder is $O_{J,K}(N^{-(J+1)})$, since $\sum_{r \geq 0} p(r)(1+r)^{2J+2} < \infty$. On the complementary range, use Lemma 4.9, which in particular gives $|\rho_N(r) - 1| \leq B$. Together with the Poisson tail estimate $\Pr\{Y > R_N\} = O(1/R_N!)$, it makes the tail smaller than every fixed power of N^{-1} for K large enough. Comparing the coefficients of $N^{-1}, \dots, N^{-J}$ gives the claim. $\square$

Proposition 5.10 (All-order near-threshold coefficient). *For each fixed $j \geq 1$, let $C_j(x)$ be the coefficient of N^{-j} obtained by expanding the probability ratio (28) and summing it against h_z , with $z = 1/L$. Then*

$$C_j(x) \sim \frac{(-1)^j (2j-1)!}{2^j j!} \frac{x}{e(L-1)^{2j}} \quad (L \downarrow 1).$$

In particular, C_j has a pole of exact order $2j$ at $L = 1$.

Proof. Write the probability ratio in the form

$$\rho_N(r) = \exp\left(\sum_{q \geq 1} \ell_q(r) N^{-q}\right), \quad \deg \ell_q = q + 1.$$

A monomial $\prod_{i=1}^d \ell_{q_i}(r)$ contributing at order N^{-j} has $\sum_i q_i = j$ and degree at most $\sum_i (q_i + 1) = j + d \leq 2j$. Equality requires $d = j$ and every $q_i = 1$. Consequently, the degree- $2j$ term comes only from $\ell_1(r)^j / j! = \alpha_r^j / j!$ and has leading coefficient $(-1/2)^j / j! = \kappa_j / (2j-1)!$.

Pass to the falling-factorial basis. For $k \geq 1$,

$$m! \sum_{r \geq m+1} p(r) r^{\underline{k}} = \frac{m!}{e} \sum_{n \geq m+1-k} \frac{1}{n!} = \frac{1}{e} m^{\underline{k-1}} + O_k(m^{k-2}).$$

Thus only the $r^{\underline{2j}}$ term reaches degree $2j-1$ in m , and its leading coefficient after taking the Poisson tail is $\kappa_j / [e(2j-1)!]$; it cannot cancel. By Lemma 5.9, the expectation of each ratio coefficient under the Poisson law vanishes. Equivalently, for any fixed order this can be checked directly in the falling-factorial basis, since $\mathbb{E}[Y^{\underline{k}}] = 1$ for $Y \sim \text{Poisson}(1)$. Thus the constant incomplete-Poisson-tail contributions cancel at every order. Since

$$\sum_{m \geq 0} m^{2j-1} z^{m+1} \sim \frac{(2j-1)!}{(1-z)^{2j}} \quad (z \uparrow 1),$$

the factorial $(2j-1)!$ cancels the denominator above, and substituting $z = 1/L$ gives the stated coefficient $\kappa_j x / [e(L-1)^{2j}]$. The exact pole order does not depend on the normalization cancellation: without it, the constant tail would satisfy $m! \Pr\{Y \geq m+1\} \asymp 1/m$ and would contribute only $O(\log(1/(L-1)))$, subordinate to $(L-1)^{-2j}$. $\square$

Remark 5.11 (A third-order consistency check). At $j = 3$ the polynomial ratio coefficient is

$$A_3(r) = -\frac{r^6}{48} + \frac{19r^5}{48} - \frac{21r^4}{8} + \frac{355r^3}{48} - \frac{17r^2}{2} + \frac{47r}{16} + \frac{5}{48}.$$

Direct evaluation in the falling-factorial basis gives $\mathbb{E}A_3(Y) = 0$, providing a further instance of Lemma 5.9. Its leading coefficient is $-1/48 = \kappa_3/5!$, since $\kappa_3 = -5/2$, exactly as Proposition 5.10 predicts.

Lemma 5.12 (All-orders likelihood-ratio remainder). *For each fixed integer $J \geq 0$, there is a constant $B_J > 0$ such that, for every $N \geq 2$ and $r \geq 0$,*

$$\left| \rho_N(r) - 1 - \sum_{j=1}^J \frac{A_j(r)}{N^j} \right| \leq B_J \frac{(1+r)^{2J+2}}{N^{J+1}}, \quad (37)$$

where the sum is empty when $J = 0$.

Proof. First suppose $r^2 \leq N$. For $N \geq 4$, expand the logarithm of the exact ratio (28) through the order needed to retain powers $N^{-1}, \dots, N^{-J}$. Taylor's theorem, summed over the r falling-product terms and the final binomial term, gives

$$\log \rho_N(r) = \sum_{j=1}^J \frac{\ell_j(r)}{N^j} + E_{J,N}(r), \quad |E_{J,N}(r)| \leq C_J \frac{(1+r)^{2J+2}}{N^{J+1}}.$$

Here $\deg \ell_j = j + 1$, and $(1+r)^2/N$ is bounded. Taylor expansion of the exponential on this bounded set, followed by collecting equal powers of N^{-1} , therefore yields (37); its polynomial coefficients are precisely the A_j .

Now suppose $r^2 > N$. The global estimate in Lemma 4.9 bounds ρ_N , while $\deg A_j \leq 2j$ gives

$$\frac{|A_j(r)|}{N^j} \leq C_j \left(\frac{(1+r)^2}{N} \right)^j.$$

Because $(1+r)^2/N > 1$, the constant term and every power through $j = J$ are bounded by a constant multiple of its $(J+1)$ st power. This proves (37) on the outer branch. The finitely many cases $N < 4$ are absorbed by increasing B_J . $\square$

Theorem 5.13 (All-orders uniform diagonal remainder). *For every fixed integer $J \geq 0$, there is a constant $A_J > 0$ such that, for all $N \geq 2$ and $x > e$, with $L = \log x$,*

$$T_N(x; N) - D(x) = \sum_{j=1}^J \frac{C_j(x)}{N^j} + R_J(N, x), \quad \boxed{|R_J(N, x)| \leq \frac{A_J x L^{2J+1}}{e N^{J+1} (L-1)^{2J+2}}.} \quad (38)$$

In particular, uniformly on every bounded interval $1 < L \leq L_0$, the remainder bound has the near-threshold pole order predicted by the formal expansion. The case $J = 0$ recovers the shape of Theorem 5.3.

Proof. Proposition 4.7, the definition of C_j , and an interchange of absolutely convergent sums give

$$\frac{R_J(N, x)}{x} = \sum_{m \geq 0} \frac{m!}{L^{m+1}} \sum_{r > m} p(r) \left(\rho_N(r) - 1 - \sum_{j=1}^J \frac{A_j(r)}{N^j} \right).$$

By Lemma 5.12, its absolute value is at most

$$\frac{B_J}{N^{J+1}} \sum_{m \geq 0} \frac{m!}{L^{m+1}} \sum_{r > m} p(r) (1+r)^{2J+2}.$$

Expanding $(1+r)^{2J+2}$ in falling factorials and using, for $k \geq 1$,

$$m! \sum_{r > m} p(r) r^{\underline{k}} = \frac{1}{e} m^{\underline{k-1}} + O_k(m^{k-2})$$

while treating the smaller constant term directly, shows that the inner factorially weighted tail is at most $C_J(1+m)^{2J+1}/e$. Put $q = 1/L$ and $p = 2J + 1$. Since $n^p \leq p! \binom{n+p-1}{p}$,

$$\sum_{m \geq 0} \frac{(1+m)^p}{L^{m+1}} = \sum_{n \geq 1} n^p q^n \leq \frac{p!q}{(1-q)^{p+1}} = \frac{p!L^p}{(L-1)^{p+1}}.$$

Combining these estimates and absorbing the J -dependent constants proves (38). $\square$

Proposition 5.14 (Borel sum of the leading-pole series). *Let*

$$\kappa_j := \frac{(-1)^j (2j-1)!}{2^j j!}, \quad \Phi(u) := \sum_{j \geq 1} \kappa_j u^j.$$

The formal series Φ is Gevrey-1. Its Borel transform and positive-direction Borel sum are

$$\widehat{\Phi}(t) := \sum_{j \geq 1} \frac{\kappa_j}{j!} t^j = \log \frac{2}{1 + \sqrt{1+2t}}, \quad (39)$$

$$\mathcal{B}\Phi(u) := \int_0^\infty e^{-t} \widehat{\Phi}(ut) dt = \int_0^\infty e^{-t} \log \frac{2}{1 + \sqrt{1+2ut}} dt, \quad u > 0. \quad (40)$$

It also has the exact Laplace-type representation

$$\boxed{\mathcal{B}\Phi(u) = \Psi(u) := \int_0^\infty (e^{-u\rho^2/2} - 1) e^{-\rho} \frac{d\rho}{\rho}.} \quad (41)$$

The transform is analytic on $\mathbb{C} \setminus (-\infty, -1/2]$ and has a square-root branch point at $t = -1/2$; hence the model series is Borel summable along the positive axis. Moreover, for every $u > 0$ one has the exact, absolutely convergent expansion

$$\boxed{\mathcal{B}\Phi(u) = -\frac{1}{2} \log u + \frac{\log 2 + \gamma}{2} + \sum_{k \geq 1} \frac{(-1)^k \Gamma(k/2)}{2k!} \left(\frac{2}{u}\right)^{k/2}.} \quad (42)$$

In particular, separating its first four terms gives the exact identity

$$\mathcal{B}\Phi(u) = -\frac{1}{2} \log u + \frac{\log 2 + \gamma}{2} - \sqrt{\frac{\pi}{2u}} + \frac{1}{2u} - \frac{1}{6} \sqrt{\frac{\pi}{2}} u^{-3/2} + \frac{1}{12u^2} + \sum_{k \geq 5} \frac{(-1)^k \Gamma(k/2)}{2k!} \left(\frac{2}{u}\right)^{k/2}.$$

Proof. Since

$$\frac{\kappa_j}{j!} = \frac{(-1)^j}{2j} \binom{2j}{j} 2^{-j},$$

the classical generating identity

$$\sum_{j \geq 1} \frac{1}{j} \binom{2j}{j} w^j = -2 \log \frac{1 + \sqrt{1-4w}}{2}$$

with $w = -t/2$ gives (39). Its principal branch has the stated cut and no singularity on the positive axis. The logarithmic growth of $\widehat{\Phi}(ut)$ at infinity makes the Laplace integral (40) convergent. Finally, for each fixed $t > 0$,

$$\widehat{\Phi}(ut) = -\frac{1}{2} \log u + \frac{1}{2} \log 2 - \frac{1}{2} \log t + o(1).$$

Integrating against e^{-t} and using $\int_0^\infty e^{-t} \log t \, dt = -\gamma$ determines the constant term; a split at $t = 1/u$ supplies a standard integrable logarithmic majorant. To obtain the lower-order terms, differentiate (40) under the integral sign. With $s = \sqrt{1 + 2ut}$, one has

$$\partial_u \log \frac{2}{1+s} = -\frac{t}{s(1+s)}, \quad ut = \frac{(s-1)(s+1)}{2},$$

so the differentiated integrand collapses to one half of $1 - s^{-1}$. Consequently,

$$\begin{aligned} u(\mathcal{B}\Phi)'(u) &= -\frac{1}{2} + \frac{1}{2} \int_0^\infty \frac{e^{-t}}{\sqrt{1+2ut}} \, dt \\ &= -\frac{1}{2} + \frac{1}{2} \sqrt{\frac{\pi}{2u}} e^{1/(2u)} \operatorname{erfc}\left(\frac{1}{\sqrt{2u}}\right). \end{aligned}$$

The second equality follows from $1 + 2ut = v^2$, which converts the remaining integral into a Gaussian tail. On the other hand, differentiation under the integral in (41), followed by one integration by parts, gives

$$\begin{aligned} u\Psi'(u) &= -\frac{u}{2} \int_0^\infty \rho e^{-u\rho^2/2-\rho} \, d\rho \\ &= -\frac{1}{2} + \frac{1}{2} \int_0^\infty e^{-u\rho^2/2-\rho} \, d\rho \\ &= -\frac{1}{2} + \frac{1}{2} \sqrt{\frac{\pi}{2u}} e^{1/(2u)} \operatorname{erfc}\left(\frac{1}{\sqrt{2u}}\right). \end{aligned}$$

Thus Ψ and $\mathcal{B}\Phi$ have the same derivative on $u > 0$; dominated convergence gives $\Psi(u) \rightarrow 0$ and $\mathcal{B}\Phi(u) \rightarrow 0$ as $u \downarrow 0$, proving their exact equality. For a second derivation of (42), take the Mellin transform of (41). For $-1 < \Re s < 0$, Fubini's theorem and the scaling $v = u\rho^2/2$ give

$$\int_0^\infty u^{s-1} \Psi(u) \, du = 2^s \Gamma(s) \Gamma(-2s).$$

Hence Mellin inversion yields

$$\mathcal{B}\Phi(u) = \frac{1}{2\pi i} \int_{(c)} \Gamma(s) \Gamma(-2s) \left(\frac{2}{u}\right)^s \, ds, \quad -1 < c < 0.$$

The residues to the left generate the divergent formal series $\Phi(u)$. To the right, the double pole at $s = 0$ gives $-\frac{1}{2} \log u + (\log 2 + \gamma)/2$, while the poles $s = k/2$, $k \geq 1$, give the summands in (42). Their absolute convergence for every $u > 0$ follows from the superexponential decay of $\Gamma(k/2)/k!$. Equivalently, this convergence can be obtained without contour shifting by using the entire expansion

$$e^{\varepsilon^2} \operatorname{erfc}(\varepsilon) = \sum_{n \geq 0} \frac{(-\varepsilon)^n}{\Gamma(n/2 + 1)}, \quad \varepsilon = (2u)^{-1/2},$$

in the exact derivative identity above and integrating locally uniformly. The preceding large- u logarithmic comparison fixes the integration constant. More explicitly, termwise integration with $k = n + 1$ first gives the coefficient

$$\frac{\sqrt{\pi} 2^{-k}}{k \Gamma((k+1)/2)}$$

multiplying $(-1)^k (2/u)^{k/2}$. Legendre's duplication formula

$$\Gamma(k/2) \Gamma((k+1)/2) = 2^{1-k} \sqrt{\pi} \Gamma(k)$$

converts it exactly to $\Gamma(k/2)/(2k!)$, the coefficient in (42). Differentiating first also incorporates the small- t region exactly rather than introducing a nonintegrable $t^{-3/2}$ term by expanding the original Laplace integrand. $\square$

Remark 5.15 (Near-threshold scaling). Proposition 5.10 shows that the operative near-threshold expansion parameter is

$$\frac{1}{N(L-1)^2},$$

not $1/N$ alone. The fixed- x expansion is uniform on every region $L \geq 1 + \delta$, but near $L = 1$ one needs $N \gg (L-1)^{-2}$ before the first-order approximation is reliable.

The singular leading factors may be written

$$\frac{(2j-1)!}{2^j j!} = \frac{(2j-1)!!}{2j} \sim \frac{j!}{2j\sqrt{\pi j}} 2^j.$$

Proposition 5.14 therefore sums the leading sequence of near-threshold pole coefficients exactly. It does not assert Borel summability of the original fixed- L error series: Proposition 5.10 takes $L \downarrow 1$ for each fixed j and supplies no large- j control at fixed L .

Theorem 5.16 (Double-scaling limit). *Let $L = 1 + \delta$, $x = e^L$, and suppose that $N \rightarrow \infty$, $\delta \downarrow 0$, and*

$$\frac{1}{N\delta^2} \rightarrow u \in (0, \infty).$$

Then

$$\boxed{\frac{e}{x}(T_N(x; N) - D(x)) \rightarrow \mathcal{B}\Phi(u),} \quad (43)$$

where $\mathcal{B}\Phi$ is equivalently given by (40) or (41).

Proof. Put $z = (1 + \delta)^{-1}$, let p_N, p be as in the proof of Theorem 5.6, and define $\rho_N(r) = p_N(r)/p(r)$, taking $\rho_N(r) = 0$ beyond the support of p_N . The exact expectation formula (27) gives

$$\frac{e}{x}(T_N(x; N) - D(x)) = \sum_{r \geq 1} \frac{h_z(r)}{r!} (\rho_N(r) - 1). \quad (44)$$

We first control the moment factor uniformly as $z \uparrow 1$. Write

$$Q_z(r) := \frac{h_z(r)}{(r-1)!z^r} = \sum_{i=0}^{r-1} \frac{z^{-i}}{(r-1)(r-2)\cdots(r-i)},$$

where the $i = 0$ denominator is the empty product. The recurrence

$$Q_z(1) = 1, \quad Q_z(r) = 1 + \frac{Q_z(r-1)}{z(r-1)}$$

shows that

$$1 \leq Q_z(r) \leq 4 \quad (r \geq 1, z \in [1/2, 1]). \quad (45)$$

Indeed, $Q_z(2) \leq 3$, $Q_z(3) \leq 4$, and the bound propagates for $r \geq 4$ because $1 + 8/(r-1) \leq 4$. For $2 \leq i \leq r/2$, the i th summand above is at most $(4/r)^i$. For $i > r/2$, it is at most $2^r/[r/2]!$; even after summing at most r such terms, this is $O(r^{-2})$. Separating the terms $i = 0, 1$ therefore gives, uniformly on this interval of z ,

$$Q_z(r) = 1 + \frac{1}{z(r-1)} + O(r^{-2}) = 1 + O(r^{-1}). \quad (46)$$

Next we control the probability ratio. For $2 \leq r \leq N/2$, the exact formula (28) may be written

$$\log \rho_N(r) = 1 + \sum_{s=0}^{r-1} \log \left(1 + \frac{1-s}{N} \right) + (N+1-r) \log \left(1 - \frac{1}{N} \right). \quad (47)$$

Using $\log(1 + y) = y - y^2/2 + \theta$ with $|\theta| \leq |y|^3$ for $|y| \leq 1/2$, and collecting the linear, quadratic, and remainder terms, yields an absolute constant C such that

$$\left| \log \rho_N(r) + \frac{r^2}{2N} \right| \leq C \left(\frac{r}{N} + \frac{r^3}{N^2} \right) \quad (2 \leq r \leq N/2). \quad (48)$$

The case $r = 1$ follows directly from (28). Hence, whenever $r\delta \rightarrow \rho \in (0, \infty)$,

$$\rho_N(r) \rightarrow e^{-u\rho^2/2}. \quad (49)$$

For domination, Lemma 4.9 gives, after renaming its absolute constant,

$$|\rho_N(r) - 1| \leq C_1 \min \left(1, \frac{r^2}{N} \right) \quad (r \geq 1). \quad (50)$$

Set $\rho = r\delta$. Since $\log(1 + \delta) \geq \delta/(1 + \delta) \geq \delta/2$ for $0 < \delta \leq 1$,

$$z^r \leq e^{-\rho/2}.$$

Equations (45) and (50), together with the boundedness of $1/(N\delta^2)$, therefore dominate the step density associated with (44) by

$$C_2 e^{-\rho/2} \min \left(u_0 \rho, \frac{1}{\rho} \right), \quad (51)$$

where $u_0 > u$ is fixed for all sufficiently small δ . This function is integrable on $(0, \infty)$.

Finally, if $r\delta \rightarrow \rho > 0$, then (46), $r \log(1 + \delta) \rightarrow \rho$, and (49) give the pointwise step-density limit

$$\frac{Q_z(r)z^r}{\rho} (\rho_N(r) - 1) \rightarrow \frac{e^{-\rho}}{\rho} (e^{-u\rho^2/2} - 1).$$

To make the last passage explicit, truncate first to $\rho \in [\varepsilon, M]$. The preceding estimates give uniform convergence there, so ordinary mesh- δ Riemann-sum convergence applies. The integrable majorant (51) makes the contributions from $(0, \varepsilon)$ and (M, ∞) uniformly small as $\varepsilon \downarrow 0$ and $M \rightarrow \infty$. Hence

$$\lim_x \frac{e}{x} (T_N(x; N) - D(x)) = \int_0^\infty (e^{-u\rho^2/2} - 1) e^{-\rho} \frac{d\rho}{\rho} = \mathcal{B}\Phi(u),$$

the last equality being (41). □

Remark 5.17 (Quantitative refinement). The proof establishes the limit through compact convergence and an integrable tail majorant; it does not assert a rate. On every fixed interval $\rho \in [\varepsilon, M]$, the Riemann discretization, (46), and (48) each produce an $O_{u,\varepsilon,M}(\delta)$ correction. The numerical first-order behavior below suggests that a quantified tail estimate should recover a global $O_u(\delta)$ rate. The same behavior persists for the moving-window theorem. An independent 80-digit evaluation using the exact recurrence reported in the supplementary reproducibility note, at $\lambda = 2, 1/2$ and $u = 1, 4$, gives successive residual-ratio pairs

$$(0.4890, 0.4940), \quad (0.4646, 0.4826), \quad (0.4911, 0.4953), \quad (0.4768, 0.4873),$$

respectively, as δ is halved from 0.1 to 0.05 to 0.025. This supports, but does not prove, a global $O_{u,\lambda}(\delta)$ rate.

Theorem 5.13 now supplies a complementary global all-orders estimate. Its factor L^{2J+1} is necessary away from the threshold, while on bounded L -intervals it has the correct $(L - 1)^{-2J-2}$ pole order predicted by the natural variable u .

Theorem 5.18 (Moving-window double scaling). *Fix $\lambda > 0$, let*

$$D_\lambda(x) := \text{Li}(x) - \text{Li}(xe^{-1/\lambda}),$$

and put $L = 1/\lambda + \delta$. If $N \rightarrow \infty$, $\delta \downarrow 0$, and

$$\frac{1}{\lambda^2 N \delta^2} \rightarrow u \in (0, \infty),$$

then

$$\frac{e^{1/\lambda}}{x} (T_N(x; \lambda N) - D_\lambda(x)) \rightarrow \mathcal{B}\Phi(u).$$

Proof. Write $\mu = 1/\lambda$, $z = L^{-1}$, and let

$$X_N \sim \text{Bin}(N+1, \mu/N), \quad Y \sim \text{Poisson}(\mu).$$

The same negative-binomial duality as in Proposition 4.7 gives $c_{N,m}^{(\lambda)} = m! \Pr\{X_N \geq m+1\}$, while $\gamma(m+1, \mu) = m! \Pr\{Y \geq m+1\}$. If $p_{N,\mu}$ and p_μ denote their masses and $\rho_{N,\mu} = p_{N,\mu}/p_\mu$ (extended by zero beyond the binomial support), then the exact analogue of (44) is

$$\frac{e^\mu}{x} (T_N(x; \lambda N) - D_\lambda(x)) = \sum_{r \geq 1} \frac{\mu^r h_z(r)}{r!} (\rho_{N,\mu}(r) - 1). \quad (52)$$

Only constants change in the proof of Theorem 5.16. Indeed,

$$\log \rho_{N,\mu}(r) = \mu + \sum_{s=0}^{r-1} \log \left(1 + \frac{1-s}{N} \right) + (N+1-r) \log \left(1 - \frac{\mu}{N} \right).$$

For fixed μ and $2 \leq r \leq N/2$, the same two-sided logarithm estimate used in (48) yields

$$\left| \log \rho_{N,\mu}(r) + \frac{r^2}{2N} \right| \leq C_\lambda \left(\frac{r}{N} + \frac{r^3}{N^2} \right). \quad (53)$$

The coefficient of N^{-1} before the error is $r - r(r-1)/2 + \mu(r-1) - \mu^2/2$; hence its quadratic part is $-r^2/2$, independently of λ . With $\rho = \lambda r \delta$ and $1/(\lambda^2 N \delta^2) \rightarrow u$, equation (53) therefore gives

$$\rho_{N,\mu}(r) \rightarrow e^{-u\rho^2/2} \quad (\lambda r \delta \rightarrow \rho > 0).$$

For the moment factor, the recursion for $Q_z(r)$ in the proof of Theorem 5.16 applies with $z \rightarrow \lambda$. Thus, for z in a fixed neighborhood of λ ,

$$Q_z(r) \leq C_\lambda, \quad Q_z(r) = 1 + O_\lambda(r^{-1}).$$

Moreover,

$$\frac{\mu^r h_z(r)}{r!} = \frac{Q_z(r)(\mu z)^r}{r}, \quad \mu z = \frac{\mu}{\mu + \delta}.$$

Since $1/r = \lambda \delta / \rho$, the right side of (52) is a mesh- $\lambda \delta$ Riemann sum. Pointwise, $(\mu z)^r \rightarrow e^{-\rho}$.

For completeness, the global comparison used for domination is

$$\rho_{N,\mu}(r) \leq e^\mu (1 + 1/N).$$

Combining it with (53), exactly as in (50), gives

$$|\rho_{N,\mu}(r) - 1| \leq C_\lambda \min\left(1, \frac{r^2}{N}\right).$$

For all sufficiently small δ , the inequality $\log(1 + \lambda\delta) \geq \lambda\delta/2$ yields $(\mu z)^r \leq e^{-\rho/2}$. Hence the step densities are dominated, for a fixed upper bound u_0 on $1/(\lambda^2 N \delta^2)$, by

$$C_\lambda e^{-\rho/2} \min\left(u_0 \rho, \frac{1}{\rho}\right),$$

which is integrable on $(0, \infty)$. The compact-plus-tail argument from Theorem 5.16 now gives

$$\lim_{x \rightarrow \infty} \frac{e^{1/\lambda}}{x} (T_N(x; \lambda N) - D_\lambda(x)) = \int_0^\infty (e^{-u\rho^2/2} - 1) e^{-\rho} \frac{d\rho}{\rho} = \mathcal{B}\Phi(u).$$

The binomial support causes no additional restriction: on compact ρ -windows, $r \asymp \delta^{-1}$ while $N\delta \rightarrow \infty$. $\square$

Theorem 5.19 (Vanishing- λ double scaling). *Fix $\delta > 0$. Let $\mu_N \rightarrow \infty$, put*

$$\begin{aligned} \lambda_N &= \frac{1}{\mu_N}, & L_N &= \mu_N + \delta, \\ x_N &= e^{L_N}, & a_N &= \lambda_N N. \end{aligned}$$

and suppose that

$$\frac{\mu_N^2}{N\delta^2} \longrightarrow u \in (0, \infty).$$

Then, with $D_\lambda(x) = \text{Li}(x) - \text{Li}(xe^{-1/\lambda})$,

$$\boxed{\frac{e^{\mu_N}}{x_N} (T_N(x_N; a_N) - D_{\lambda_N}(x_N)) \longrightarrow \Psi_\delta(u),} \quad (54)$$

where

$$\boxed{\Psi_\delta(u) := \int_0^\infty e^{-\rho} \frac{e^{-u(\rho-\delta)^2/2} - 1}{\rho - \delta} d\rho.} \quad (55)$$

The integrand is understood by continuity at $\rho = \delta$. In particular, for the anchored choice $\delta = \log 2$ one has $D_{\lambda_N}(x_N) = \text{Li}(x_N)$ under the normalization (2).

Proof. Write $\mu = \mu_N$, $L = \mu + \delta$, and $z = L^{-1}$. With $X_N \sim \text{Bin}(N+1, \mu/N)$ and $Y \sim \text{Poisson}(\mu)$, the expectation identity used in (52) gives

$$\frac{T_N(x_N; a_N) - D_{\lambda_N}(x_N)}{x_N} = \mathbb{E} h_z(X_N) - \mathbb{E} h_z(Y).$$

The elementary gamma integral

$$h_z(r) = \sum_{m=0}^{r-1} m! z^{m+1} = \int_0^\infty e^{-Lt} \frac{1-t^r}{1-t} dt$$

is valid with the quotient interpreted continuously at $t = 1$. Taking expectations and using the probability generating functions of X_N and Y therefore yields the exact identity

$$\frac{e^\mu}{x_N} (T_N(x_N; a_N) - D_{\lambda_N}(x_N)) = \int_0^\infty e^{-\delta t} \frac{1 - e^{S_N(t)}}{1-t} dt, \quad (56)$$

where

$$S_N(t) = \mu(1-t) + (N+1) \log \left(1 + \frac{\mu(t-1)}{N} \right).$$

The assumption implies $\mu/N \rightarrow 0$, so the logarithm is defined for all $t \geq 0$ once N is large. For every fixed $t \geq 0$, Taylor expansion gives

$$S_N(t) = \frac{\mu(t-1)}{N} - \frac{(N+1)\mu^2(t-1)^2}{2N^2} + O_t \left(\frac{\mu^3}{N^2} \right) \rightarrow -\frac{u\delta^2(t-1)^2}{2}.$$

We justify passage under the integral. On a fixed neighborhood of $t = 1$, the same expansion with a uniform remainder and the boundedness of μ^2/N give $|S_N(t)| \leq C(|t-1| + |t-1|^2)$; hence $|1 - e^{S_N(t)}|/|1-t|$ is uniformly bounded there. Away from $t = 1$ on a fixed compact interval, ordinary uniform convergence applies. Finally, for $t \geq 1$, the inequality $\log(1+s) \leq s$ gives

$$S_N(t) \leq \frac{\mu(t-1)}{N}.$$

For all sufficiently large N , $\mu/N \leq \delta/2$, so the tail of the absolute integrand in (56) is bounded by a constant multiple of $e^{-\delta t}(1 + e^{\delta(t-1)/2})/(t-1)$, which is integrable away from the already treated point $t = 1$. Dominated convergence now gives

$$\lim \frac{e^\mu}{x_N} (T_N(x_N; a_N) - D_{\lambda_N}(x_N)) = \int_0^\infty e^{-\delta t} \frac{1 - e^{-u\delta^2(t-1)^2/2}}{1-t} dt.$$

The change of variables $\rho = \delta t$ is exactly (55). □

Remark 5.20 (The turning point and recovery of the Borel model). The fixed- λ proof cannot be made uniform merely by enlarging its constants. If $r/\mu \rightarrow y > 1$, the defining sum for the moment factor gives

$$Q_z(r) \rightarrow \sum_{i \geq 0} y^{-i} = \frac{y}{y-1},$$

so on $\rho = \delta r/\mu > \delta$ its step density acquires the factor $1/(\rho - \delta)$ rather than $1/\rho$. For $r/\mu < 1$, the terms in $Q_z(r)$ grow toward the opposite endpoint and no bounded- Q_z argument is available. Formula (56) treats the cancellation across that turning region exactly.

Splitting (55) at $\rho = \delta$ shows that the lower part is positive and the upper part negative. For fixed $u, \delta > 0$ the lower contribution is nonzero, and

$$0 < \int_0^\delta e^{-\rho} \frac{e^{-u(\rho-\delta)^2/2} - 1}{\rho - \delta} d\rho \leq \frac{u\delta^2}{4}.$$

Consequently it vanishes as $\delta \downarrow 0$ for each fixed u , and $\Psi_\delta(u) \rightarrow \mathcal{B}\Phi(u)$. This convergence is not uniform in u : as the next corollary shows, for large u the two sides of the turning point carry opposite logarithmic divergences whose cancellation has no analogue when the pole lies at the endpoint $\delta = 0$. At small u ,

$$\Psi_\delta(u) = -\frac{1-\delta}{2}u + O_\delta(u^2). \tag{57}$$

Thus the fixed-window Borel shape is recovered only after the absolute boundary gap also shrinks, with the order of the $u \rightarrow \infty$ and $\delta \downarrow 0$ limits being essential.

Theorem 5.21 (Exact two-parameter crossover). *For $u, \delta > 0$, put $v = \delta\sqrt{u}$ and define*

$$\mathcal{C}_u(v) := \int_0^v e^{t/\sqrt{u}} \frac{1 - e^{-t^2/2}}{t} dt. \tag{58}$$

Then the shifted model satisfies the exact identity

$$\boxed{\Psi_\delta(u) = e^{-\delta} [\mathcal{B}\Phi(u) + \mathcal{C}_u(\delta\sqrt{u})]}. \quad (59)$$

For every $u, \delta > 0$ one also has the global squeeze

$$\mathcal{C}(\delta\sqrt{u}) \leq \mathcal{C}_u(\delta\sqrt{u}) \leq e^\delta \mathcal{C}(\delta\sqrt{u}), \quad (60)$$

and hence the quantitative approximation

$$0 \leq \Psi_\delta(u) - e^{-\delta} [\mathcal{B}\Phi(u) + \mathcal{C}(\delta\sqrt{u})] \leq (1 - e^{-\delta}) \mathcal{C}(\delta\sqrt{u}) \leq \delta \mathcal{C}(\delta\sqrt{u}). \quad (61)$$

If $u \rightarrow \infty$, $\delta \downarrow 0$, and $\delta\sqrt{u} \rightarrow v \in [0, \infty)$, then

$$\Psi_\delta(u) - \mathcal{B}\Phi(u) \rightarrow \mathcal{C}(v), \quad \mathcal{C}(v) := \int_0^v \frac{1 - e^{-t^2/2}}{t} dt, \quad (62)$$

uniformly for v in compact subsets of $[0, \infty)$. The crossover function has the closed form

$$\boxed{\mathcal{C}(v) = \frac{1}{2} \left[\gamma + \log\left(\frac{v^2}{2}\right) + E_1\left(\frac{v^2}{2}\right) \right]}, \quad v > 0, \quad (63)$$

with $\mathcal{C}(0) = 0$, and satisfies

$$\mathcal{C}(v) = \frac{v^2}{4} + O(v^4) \quad (v \downarrow 0), \quad \mathcal{C}(v) = \log v + \frac{\gamma - \log 2}{2} + O\left(\frac{e^{-v^2/2}}{v^2}\right) \quad (v \rightarrow \infty).$$

More generally, if $\delta \downarrow 0$, $u \rightarrow \infty$, and $\delta \log u \rightarrow 0$, with no restriction on the limit of $\delta\sqrt{u}$, then

$$\boxed{\Psi_\delta(u) - \mathcal{B}\Phi(u) - \mathcal{C}(\delta\sqrt{u}) \rightarrow 0}. \quad (64)$$

Thus $v^2 = u\delta^2$ is the governing crossover coordinate in the small- δ transition between the endpoint-pole Borel model and the interior-pole shifted phase.

Proof. In (55), set $q = \rho - \delta$ and then $s = \sqrt{u}q$. The apparent singularity at $s = 0$ is removable, and splitting the resulting integral there gives

$$\Psi_\delta(u) = e^{-\delta} \left[\int_0^\infty e^{-s/\sqrt{u}} \frac{e^{-s^2/2} - 1}{s} ds + \int_0^{\delta\sqrt{u}} e^{t/\sqrt{u}} \frac{1 - e^{-t^2/2}}{t} dt \right].$$

The first integral is $\mathcal{B}\Phi(u)$ after $s = \sqrt{u}\rho$, proving (59). On compact v -intervals, $\mathcal{C}_u(v) \rightarrow \mathcal{C}(v)$ uniformly. Also $\mathcal{B}\Phi(u) = O(\log u)$ by (42), while $\delta = O(u^{-1/2})$ when v remains bounded. Hence $(e^{-\delta} - 1)\mathcal{B}\Phi(u) \rightarrow 0$, which proves (62).

On $0 \leq t \leq \delta\sqrt{u}$ one has $1 \leq e^{t/\sqrt{u}} \leq e^\delta$. Positivity of the remaining factor proves (60), and (61) follows from (59). To prove the extended limit, write

$$|\Psi_\delta(u) - \mathcal{B}\Phi(u) - \mathcal{C}(\delta\sqrt{u})| \leq (1 - e^{-\delta}) (|\mathcal{B}\Phi(u)| + \mathcal{C}(\delta\sqrt{u})).$$

Here $\mathcal{B}\Phi(u) = O(\log u)$, while $\mathcal{C}(\delta\sqrt{u}) = O(1 + \log u)$ when $\delta \leq 1$, by the displayed asymptotics. The hypothesis $\delta \log u \rightarrow 0$ therefore proves (64).

For the closed form, set $y = t^2/2$:

$$\mathcal{C}(v) = \frac{1}{2} \int_0^{v^2/2} \frac{1 - e^{-y}}{y} dy.$$

The standard identity for this entire combination gives (63); its small- and large- v expansions follow from those of E_1 . $\square$

Proposition 5.22 (Exact decomposition of the shifted model). *For $u, \delta > 0$,*

$$\boxed{\Psi_\delta(u) = e^{-\delta} \left[\text{Ei}(\delta) - \pi \operatorname{erfi}\left(\frac{1}{\sqrt{2u}}\right) + A_1(u, \delta) - A_2(u, \delta) \right]}, \quad (65)$$

where

$$A_1(u, \delta) := \int_\delta^\infty \frac{e^{-uq^2/2}}{q} dq = \frac{1}{2} E_1\left(\frac{u\delta^2}{2}\right) > 0, \quad (66)$$

$$A_2(u, \delta) := \int_{-\infty}^{-\delta} (e^{-q} - 1) \frac{e^{-uq^2/2}}{q} dq < 0, \quad (67)$$

and $E_1(v) = \int_v^\infty e^{-s} ds/s$. For fixed $\delta > 0$,

$$A_1(u, \delta), A_2(u, \delta) = O_\delta(e^{-u\delta^2/2}) \quad (u \rightarrow \infty).$$

Proof. Set $q = \rho - \delta$ in (55) and split $e^{-q} = 1 + (e^{-q} - 1)$. The principal value of the first part over $[-\delta, \infty)$ cancels on $[-\delta, \delta]$ and leaves A_1 . Extend the second part to the whole real line and subtract its portion on $(-\infty, -\delta)$, which is A_2 . It remains to evaluate

$$J(a) := \int_{-\infty}^\infty (e^{-aq} - 1) \frac{e^{-uq^2/2}}{q} dq, \quad 0 \leq a \leq 1.$$

Differentiation under the integral gives

$$J'(a) = - \int_{-\infty}^\infty e^{-aq - uq^2/2} dq = -\sqrt{\frac{2\pi}{u}} e^{a^2/(2u)}.$$

Since $J(0) = 0$, integration from 0 to 1 yields $J(1) = -\pi \operatorname{erfi}(1/\sqrt{2u})$. The remaining principal-value term is $-\text{PV} \int_{-\delta}^\infty e^{-q} dq/q = \text{Ei}(\delta)$, proving (65). Equation (66) follows from $s = uq^2/2$. For A_2 , put $q = -s$; once $u\delta > 1$, the exponent $-us^2/2 + s$ decreases for $s \geq \delta$. Standard endpoint bounds then give the stated exponential estimates for both tails. $\square$

Corollary 5.23 (Large- u phase distinction). *For every fixed $\delta > 0$,*

$$\boxed{\lim_{u \rightarrow \infty} \Psi_\delta(u) = e^{-\delta} \text{Ei}(\delta)}. \quad (68)$$

More precisely, for every fixed $M \geq 1$,

$$\Psi_\delta(u) = e^{-\delta} \text{Ei}(\delta) - e^{-\delta} \sqrt{2\pi} \sum_{k=0}^{M-1} \frac{u^{-k-1/2}}{2^k k! (2k+1)} + O_{M,\delta}(u^{-M-1/2} + e^{-u\delta^2/2}). \quad (69)$$

The portions of (55) below and above $\rho = \delta$ are respectively

$$\frac{e^{-\delta}}{2} \log u + O_\delta(1), \quad -\frac{e^{-\delta}}{2} \log u + O_\delta(1).$$

For $\delta = \log 2$, the limit is $\frac{1}{2} \text{li}(2) = 0.5225818901\dots$, whereas $\mathcal{B}\Phi(u) \sim -\frac{1}{2} \log u$ tends to $-\infty$.

Proof. The expansion follows from Proposition 5.22 and

$$\pi \operatorname{erfi}\left(\frac{1}{\sqrt{2u}}\right) = \sqrt{2\pi} \sum_{k \geq 0} \frac{u^{-k-1/2}}{2^k k! (2k+1)}.$$

For the separate logarithms, split (55) at $\rho = \delta$ and use $s = \sqrt{u}|\rho - \delta|$. The coefficient is one half because $\int_{u^{-1/2}}^1 dq/q = \frac{1}{2} \log u$; the signs follow from the two sides of the pole. $\square$

Remark 5.24 (One crossover, two logarithmic descriptions). The paired logarithms in Corollary 5.23 are already contained in the exact crossover identity. Indeed, putting $v = \delta\sqrt{u}$ and then $t = \sqrt{u}r$ gives

$$\mathcal{C}_u(v) - \mathcal{C}(v) = \int_0^\delta \frac{e^r - 1}{r} \left(1 - e^{-ur^2/2}\right) dr.$$

For fixed $\delta > 0$, dominated convergence therefore yields

$$\mathcal{C}_u(\delta\sqrt{u}) - \mathcal{C}(\delta\sqrt{u}) \longrightarrow \int_0^\delta \frac{e^r - 1}{r} dr = \text{Ei}(\delta) - \gamma - \log \delta = \sum_{k \geq 1} \frac{\delta^k}{k k!}.$$

At the same time, the large-argument expansions in (42) and Theorem 5.21 give

$$\mathcal{B}\Phi(u) + \mathcal{C}(\delta\sqrt{u}) \longrightarrow \gamma + \log \delta.$$

Thus the terms $\mp \frac{1}{2} \log u$ cancel inside a single identity, while the retained factor $e^{t/\sqrt{u}}$ supplies exactly the residual power series of $\text{Ei}(\delta)$. Substitution in (59) recovers $\Psi_\delta(u) \rightarrow e^{-\delta} \text{Ei}(\delta)$ directly. Proposition 5.22 resolves the same crossover into special-function tails: $A_1 - A_2$ is exponentially small when $u\delta^2 \gg 1$ and ceases to be so on the scale $u \asymp \delta^{-2}$. Conversely, as $\delta \downarrow 0$ with u fixed, the logarithms of $\text{Ei}(\delta)$ and $A_1(u, \delta)$ cancel to recover $\mathcal{B}\Phi(u)$.

Proposition 5.25 (First vanishing- λ correction). *Fix $u, \delta > 0$, let $\mu \rightarrow \infty$, and take*

$$N = \text{nint}\left(\frac{\mu^2}{u\delta^2}\right), \quad \lambda = \mu^{-1}, \quad x = e^{\mu+\delta}, \quad a = \lambda N.$$

Then

$$\frac{e^\mu}{x} (T_N(x; a) - D_\lambda(x)) = \Psi_\delta(u) + \frac{K_\delta(u)}{\mu} + O_{u,\delta}(\mu^{-2}), \quad (70)$$

where

$$K_\delta(u) = \int_0^\infty e^{-\delta t} e^{-u\delta^2(t-1)^2/2} \left[u\delta^2 + \frac{u^2\delta^4}{3}(t-1)^2 \right] dt > 0. \quad (71)$$

Thus every sufficiently large finite- μ value lies above its limiting value.

Proof. Put $q = t - 1$ and $b = u\delta^2$. Nearest-integer rounding gives $\mu^2/N = b + O_{u,\delta}(\mu^{-2})$. Expanding the exponent in (56) one order further gives, uniformly for bounded q ,

$$S_N(t) = -\frac{bq^2}{2} + \frac{1}{\mu} \left(bq + \frac{b^2q^3}{3} \right) + O_{u,\delta,q}(\mu^{-2}).$$

After exponentiation, substitution in (56), and division by $1 - t = -q$, the coefficient of μ^{-1} is exactly (71).

For a global remainder, fix any $\theta \in (0, 1)$ and split at $q = \mu^\theta$. On $0 \leq q \leq \mu^\theta$, the logarithmic remainder is $o(q^2)$ relative to the leading Gaussian exponent. After exponentiation it is therefore dominated by $e^{-bq^2/4}$ times $C_{u,\delta}(1 + |q|^4)\mu^{-2}$, an integrable majorant whose integral is $O_{u,\delta}(\mu^{-2})$. On $q \geq \mu^\theta$, the inequality $\log(1 + s) \leq s - s^2/[2(1 + s)]$ first gives Gaussian suppression $O(e^{-c_{u,\delta}\mu^{2\theta}})$ while $s = \mu q/N$ remains bounded; once s is large, the outer factor $e^{-\delta t}$ and $\mu/N < \delta/2$ give an exponential bound in q . Thus the discarded region is smaller than every power of μ^{-1} , independently of a Taylor-remainder constant. This proves the stated $O_{u,\delta}(\mu^{-2})$ remainder. Positivity of K_δ is immediate from (71). $\square$

6 Consequences and limitations

6.1 What the coupled theorem does establish

The diagonal choice $a = N$ is not a heuristic substitution into a divergent series. After the substitution is made at finite N , exact algebra converts the double sum into positive coefficients $c_{N,m}$. Those coefficients converge to incomplete-gamma moments, are uniformly dominated, and satisfy an explicit global estimate. Therefore the conclusion is a conventional theorem about a sequence of real numbers.

The theorem also explains the restriction $x > e$. The coefficients tend to $\gamma(m+1, 1) \sim e^{-1}/(m+1)$, so the limiting generating series has radius one in the variable $1/L$. The boundary $L = 1$ is harmonic, and there is no absolute or conditional rescue because the coefficients are positive.

6.2 Anchoring and what the unit window does not establish

The unit-window result alone does not isolate $\text{Li}(x)$ from the single difference $D(x) = \text{Li}(x) - \text{Li}(x/e)$. The identity

$$\text{Li}(x) = D(x) + \text{Li}(x/e)$$

is exact, but a formula for this one difference determines only the gap between two values of Li . More precisely, if $F(x)$ satisfies

$$F(x) - F(x/e) = D(x),$$

then so does $F(x) + H(x)$ for every function H invariant under the argument map, $H(x) = H(x/e)$. In logarithmic coordinates, such an H may be any period-one function of $\log x$. Thus two convergent series for the same fixed window are dependent identities; by themselves they do not determine the additive normalization.

Proposition 4.5 changes this conclusion once the window length is allowed to depend on x . For $x > 2$, take

$$\lambda = \frac{1}{\log x - \log 2}, \quad a = \lambda N.$$

Then $1/\lambda = \log x - \log 2$ and the moving-window limit becomes

$$T_N(x; \lambda N) \longrightarrow \text{Li}(x) - \text{Li}(2) = \text{Li}(x),$$

under the offset convention of this paper. Therefore the triangular diagonal family can be anchored directly by spanning the whole interval from 2 to x .

The cost of this direct anchoring is poor conditioning at large x . Its effective boundary ratio $1 - \log 2/L$ tends to one as $x \rightarrow \infty$. A separate multiscale construction that repairs this computational defect, together with certified enclosures and parameter tuning, is developed in the supplementary note; it is not needed for the transition theorems proved here.

The vanishing- λ scaling theorem makes the conditioning quantitative. Its natural dimensionless parameter is

$$u = \frac{1}{N(\lambda L - 1)^2}.$$

For the anchored choice $\lambda = (L - \log 2)^{-1}$ this becomes

$$u = \frac{1}{N} \left(\frac{L - \log 2}{\log 2} \right)^2. \tag{72}$$

Theorem 5.19, with $\delta = \log 2$ and $\mu = L - \log 2$, proves that a nontrivial transition occurs precisely when the squared factor in (72) is comparable to N . Thus the perturbative regime lies beyond the scale

$N \asymp ((L - \log 2)/\log 2)^2$, which grows like $(\log_2 x)^2$; at $x = 10^6$ the squared factor is approximately 358. The limiting shape is $\Psi_{\log 2}$, not the fixed-window model $\mathcal{B}\Phi$. This quadratic deterioration is an external consequence of the scaling theory: it explains why the natural directly anchored member requires truncation orders growing like $(\log x)^2$.

Nor does the diagonal limit assign a canonical sum to the divergent fixed- a outer series. Such an interpretation would require a specified summability method and proof that its value agrees with the diagonal limit. No such assertion is needed here.

7 Conclusion

The triangular family (3) diverges for every fixed parameter, but its finite diagonal sequence $a = N$ has both Riemann-sum and probability-tail structures. These give the exact limit $\text{Li}(x) - \text{Li}(x/e)$ for $x > e$, divergence at and below the sharp boundary, explicit global estimates, the first two closed-form error coefficients, and an all-orders uniform remainder with the correct near-threshold pole.

The degeneration of this fixed- x expansion is itself structured. The diagonal and every fixed moving window share the Borel model $\mathcal{B}\Phi$ after the natural threshold rescaling. When the window length diverges, the limit changes to the shifted model Ψ_δ and places the poor conditioning of the directly anchored member on the rigorous $N \asymp (\log x)^2$ scale. The exact crossover identity and its global squeeze then identify $v^2 = u\delta^2$ as the coordinate connecting the endpoint and interior-turning-point phases. The special-function decomposition, finite large- u limit, paired logarithmic cancellation, and explicit positive $1/\mu$ correction make this a genuine phase distinction rather than a uniform perturbation.

The analytic ingredients— incomplete-gamma identities, Poisson approximation, Borel–Laplace summation, Mellin inversion, and error-function asymptotics—are classical. The contribution is their precise synthesis for the specific array T_N , not a new general summability method. Natural next questions are to sharpen constants in the all-orders remainder, obtain global fixed-window transition rates, and identify further probability-generating-function representations. Auxiliary finite-window enclosures, multiscale formulas, numerical evidence, and implementation details are retained in the accompanying supplementary note.

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